

CEO Replacements and Firm Growth: Correcting for Small-Sample Bias in Fixed-Effect Estimates

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ABSTRACT

We develop a placebo-controlled method to correct small-sample bias in firm–CEO two way fixed-effects estimates. The approach exploits within-firm CEO replacements and measures changes in CEO quality from differences in firm performance across adjacent CEO spells. Placebo transitions are assigned to matched firms without CEO changes, allowing us to identify and remove bias in second moments and event-study estimates. Monte Carlo simulations show that the method accurately recovers true variances, covariances, and dynamic effects in short panels. Applying the method to the universe of Hungarian firms, we find that conventional estimators substantially overstate the importance of CEOs and generate spurious pre-trends before CEO transitions.

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1 INTRODUCTION

The role of managers in shaping firm outcomes has attracted substantial attention in economics, finance, and management. Because researchers rarely observe the managerial traits relevant for firm behavior, the dominant empirical approach studies the role of managers by capturing managerial heterogeneity through firm and manager fixed effects. These studies find that managers account for a substantial share of variation in firm outcomes.

A central challenge of the two-way fixed effects estimation is that managerial spells in firms are usually short. As a consequence, estimated manager effects are noisy, and any statistic involving second moments are subject to limited mobility bias even if many CEO transitions are observed [13]. A growing methodological literature has therefore focused on correcting biases in second moments of estimated fixed effects. These include parametric corrections for estimation error [1, 10], leave-one-out estimators [14], and approaches based on grouping observations to reduce dimensionality [4, 5]. While these contributions have substantially improved inference, they rely on settings where managerial mobility across firms is sufficiently rich to identify managerial heterogeneity from observed movements across organizations.

Existing evidence on managerial effects typically concerns large, often publicly listed corporations, where managers switch firms and where rich information on executives is available (as we detail at the

end of this section).¹ Much less is known about managerial effects among small and medium-sized enterprises (SMEs), even though they account for roughly two-thirds of employment in advanced economies and an even larger fraction in developing countries [18]. In these firms managers typically spend most or all of their careers at a single firm, and little information on firm outcomes is available beyond standard accounting records. As a result, the approach used in the literature would exclude most firms and managers from the analysis, leaving open the question of how important managers are in the broader population of firms. Moreover, the estimation sample is further reduced because the standard two-way fixed effects estimation can only be applied to the connected component of the data. These requirements reduce the share of firms which can be used in the analysis making estimates imprecise due to small sample problems. The small sample also calls into question the external validity of the results if SMEs which switch CEOs are different from the bulk of firms which do not.²

This paper makes three contributions. First, we develop an estimator that recovers unbiased second moments of CEO-spell effects under mild identification assumptions and corrects the bias in variance decompositions and event-study coefficients. Second, we construct a new dataset linking CEOs to the universe of almost half million Hungarian firms for the period of 1992 – 2023. Third, we use the new dataset and methodology to quantify the importance of managerial heterogeneity in SMEs, showing that conventional estimators substantially overstate the explanatory power of CEO changes and generate spurious dynamic patterns around CEO transitions.

The starting point of our estimation is a standard two-way fixed effects model in which the dependent variable is a measure of firm performance (e.g., revenue), while firm and CEO effects capture persistent heterogeneity across firms and managers. Specifically, firm performance is modeled as the sum of a firm fixed effect, a CEO-spell fixed effect, and a transitory error term.

Because CEO mobility across firms is relatively rare among SMEs, we use within-firm CEO replacements for identification. Instead of using separate CEO fixed effects, we transform the two-way fixed-effects model into first differences, which removes the firm fixed effect from the specification and estimate the variation of firm outcomes around the CEO change. The key advantage of this approach is that it can be applied to the full set of firms experiencing

¹Even among large firms, external CEO appointments remain the exception rather than the rule. Among the listed companies of the US, 72% of CEO hires are internal promotions and over 80% are current or former employees or board members [7].

²[7] shows that among the US listed companies, external hiring depends of firm characteristics.

a CEO transition, not only those in the largest connected component of the CEO–firm network.

To correct the small-sample noise inherent in estimated CEO fixed effects, we introduce a *placebo-controlled debiasing* method. To each firm with a CEO change, we assign one or more matched control firm that does not change its CEO during the same period. We then assign the control firm a placebo CEO transition in the same year as the treated firm, replicating the treated firm’s spell structure. Because no actual CEO change occurs in the placebo firm, any estimated variance, covariance, or event-study “effect” measured around the placebo transition reflects only estimation noise arising from short CEO spells and persistent firm-level shocks. Thus, changes around the placebo CEO replacement can be used to purge the estimated effect of CEO change from the idiosyncratic noise arising from the short time series of CEO spell.³

Our estimation uses two identifying assumptions. The first is the standard *exogeneity assumption* and states that changes in CEO-spell effects are uncorrelated with firm-level shocks. Under this assumption, the estimated change in the CEO effect is an unbiased measure of the true change in CEO quality. We call the second identifying assumption the *proportional autocovariance assumption*, which states that the variance-covariance matrix of shocks is identical in the treated and placebo samples up to a scalar multiplier. This assumption requires the temporal correlation structure of idiosyncratic firm-level shocks to be the same across the treated and placebo groups. At the same time, it allows firms which replace their CEO to experience larger or smaller idiosyncratic shocks than firms with stable CEOs. Conceptually, the proportional autocovariance assumption resembles the parallel trends assumption in difference-in-differences designs. Our method can be viewed as difference-in-differences for second moments: we difference out the bias by comparing treated and control firms’ variances and covariances.

Under the two assumptions, the placebo sample reproduces the same small-sample bias present in the treated sample. Subtracting the placebo moments from the corresponding moments based on the treated firms yields unbiased estimates of CEO-related variance, covariance, and dynamic treatment effects.

We construct two statistics, which contain the second moments of the data and thus are contaminated by the small sample bias. The first is the R^2 of the estimation, which measures the share of the variance of the firm variable attributable to CEO change. The second is a full set of event-time coefficients, which shows both the trend in the dependent variable pre-transition and its post-transition dynamics. These features are particularly important because pre-trends are commonly used to assess the validity of the identification strategy, while the shape of the post-transition response reveals how quickly CEO effects materialize and whether they persist over time.

We test the validity of the placebo-controlled debiasing in Monte Carlo experiments designed to mimic the characteristics of firm-level panels with CEO transitions. The simulations generate firms with two CEO spells and outcomes determined by a CEO effect and an idiosyncratic shock process. We consider a range of empirically

relevant environments, including short CEO tenures (the baseline scenario), persistent shocks, excess variance in treated firms, all complications at the same time, long panels and a pre-trend (to check how the method performs when some of the identifying assumptions are not satisfied). The experiments show that conventional estimators substantially overstate the variance of CEO effects and can generate spurious pre-trends and distorted post-transition dynamics even when the identifying assumptions hold. In contrast, the placebo-controlled estimator consistently recovers the true variances, covariances, and event-study coefficients across all scenarios. These results demonstrate that the method is robust to all the complications we inserted in the data.

We apply the placebo-controlled debiasing method to the universe of Hungarian firms between 1992 and 2023, leveraging administrative data that link CEOs to firm balance sheets and income statements. The data cover the population of SMEs (471,000 firms and over 2.5 million firm-years), 475,000 distinct CEOs and about 266,000 within firm CEO transitions.⁴

We estimate the variation of several firm attributes around CEO transitions: output, inputs (labor and capital), and productivity (return on assets and labor productivity). The empirical results show that conventional estimators substantially overstate the importance of CEOs. The R^2 of conventional estimates is 0.65–0.75, whereas the debiased estimates imply R^2 s between 0.19–0.29. The explanatory power of the estimation is larger when the incumbent and the new CEO differ along some dimension (founder–nonfounder, female–male, age), or when one CEO cleanly replaces another, rather than cases where one CEO remains permanently with the firm while a second CEO leaves or joins. We also find that event-study estimates based on conventional methods display pronounced pre-trends that disappear after debiasing, indicating that they are artifacts of small-sample bias rather than evidence against the identification strategy.

Literature. Our work connects primarily to the literature using firm and manager fixed effects to estimate the CEO influence on firm policy. [3] was the first to use two-way fixed effects to study the importance of managers, and showed that CEOs influence investment, growth, productivity and other firm outcomes. Researchers connected managerial fixed effects with profitability [17], risk taking [21], and employee quality [16] (this paper used manager and worker fixed effects). Papers established that CEO effects increase over time [20] and are correlated with nationwide institutions, such as the origin of the legal system, labor market flexibility, and ownership structure [6]. These studies use data from listed companies. [19] uses data of large listed and private companies, and finds that that CEOs affect private firms to a larger extent. One exception in terms of firm size is [2], who use the universe of Italian limited liability companies over 20 employees. The paper finds that managerial quality is associated with the adoption of better managerial practices. Similar methods were also applied to the public sector and also find that CEOs are important in this area as well [8, 11].

³Using firms which do not change CEOs to control for noise in the data is in the spirit of [9] and [12].

⁴Perhaps the best alternative of our method is the leave-one-out method [14]. This method is based on the sample of firms which have a CEO transition where both CEOs also served in another firm. In addition, this method can be performed only on the connected component of the data. The final sample is XX firms, only XX% of the data.

In contrast to these datasets, our study uses the universe of Hungarian firms (we only drop corporations with fewer than 2 employees on average). For identification, our method uses all the firms which experience a CEO change (about a quarter of million transitions) and we do not need to restrict the sample to the connected component.

Methodologically, we connect to the literature correcting the limited mobility bias in two-way fixed effects analysis. In terms of variance-covariance correction, three broad methods are proposed: assuming and estimating a parametric model for error covariance structure [1, 10, 13, 15], leave-one-out adjustments [14], and grouping observations to reduce the number of parameters to be estimated [5]. Parametric methods require correctly specifying and estimating complex autocorrelation patterns, while our method needs only mild identifying assumptions. The leave-one-out method is impractical in CEO markets where most managers work at only one firm and mobility is limited. Since our interest is understanding heterogeneity of CEOs and factors that correlate with this heterogeneity, keeping the original dimensionality of the data is preferable.

The main difference between the methods described above and ours is the we estimate from within firm, rather than between firm, CEO changes. Similar approach was employed [9] and [12]. They state that the limited mobility bias distorts the second estimated moments of CEO fixed effects and use placebo CEO switches to correct for this bias. They do not, however, demonstrate formally under what identifying assumptions will the corrected estimate be unbiased, nor they correct for the coefficients of event time analysis. In addition, they use listed firms while we use the whole Hungarian population.

Our paper also builds on the two-way fixed effects literature in labor economics that decomposes wages into worker and firm components [13]. Unlike the labor literature, our focus is on understanding manager fixed effects rather than firm fixed effects. We include firm fixed effects in our estimation model, but remove them in the estimation step by differencing or demeaning, so we cannot comment on the correlation between firm and worker (CEO) effects, a common question in labor studies.

The paper proceeds as follows. Section 2 presents the econometric problem, explains briefly the small sample bias and shows its implication on commonly used statistics. Section 3 introduces the placebo-controlled debiasing method, which we evaluate in Section 4 with Monte Carlo estimations. Section 5 applies the method to CEO transitions in Hungarian firms and studies the role of managerial heterogeneity in shaping firm outcomes. Section 6 concludes.

2 THE ECONOMETRIC PROBLEM

Let q_{it}^k denote measure k of firm i 's performance in year t . Such measures can include log revenue, return on assets, log labor productivity, etc. For each measure k , we write

$$q_{it}^k = \alpha_i^k + z_{is}^k + e_{it}^k, \quad (1)$$

where α_i^k is a firm fixed effect, s indexes the CEO spell (first, second, etc), and z_{is}^k is the component of firm performance associated with that CEO spell. The spell component is constant during a CEO

spell and changes when the firm replaces the CEO. The remaining variation is captured by the firm-level shock e_{it}^k .

Let S_i denote the number of CEO spells observed at firm i . We impose strict exogeneity of firm-level shocks with respect to the entire CEO-spell history:

$$E[e_{it}^k \mid \alpha_i^k, \{z_{is}^k\}_{s=1,2,\dots,S_i}] = 0 \quad \text{for every } i \text{ and } t. \quad (2)$$

This assumption requires the firm-level shock in each period to have mean zero conditional on the firm's entire sequence of CEO-spell effects and CEO-spell assignments. In particular, it rules out systematic selection of CEO transition timing on realizations of the firm-level shocks. It therefore implies that the average shock within each CEO spell is zero in expectation, so that the difference between average outcomes in two adjacent spells is an unbiased measure of the change in the CEO-spell component.

The object z_{is}^k is a firm-specific CEO-spell effect, rather than a portable person effect that follows the CEO across firms. A large share of the CEOs in our comprehensive dataset are observed at only one firm, therefore we do not use mobility across firms to identify their effects. The spell component may capture the CEO's skill, the match between the CEO and the firm, or other persistent features associated with that particular CEO spell. Throughout the paper, we refer to it as a CEO effect for brevity, but the empirical object is the component of firm performance associated with a CEO spell at a particular firm.

2.1 Changes in CEO-spell components

Our interest is not in the level of the firm fixed effect, hence to remove α_i^k , we difference outcomes across adjacent CEO spells within the same firm. As we are interested in changes within the firm, we do not need to restrict our attention only to the connected CEO-firm mobility network, and we can study all firms with a CEO replacement. For firm i , consider two adjacent CEO spells, $s-1$ and s . The true change in the CEO-spell component measured using outcome k is

$$\Delta z_{is}^k = z_{is}^k - z_{i,s-1}^k. \quad (3)$$

We do not observe either spell component directly. Let T_{is} denote the number of years in which firm i is observed during CEO spell s . We estimate the difference between the two spell components using the difference between the average outcomes observed during the two spells:

$$\Delta \bar{z}_{is}^k = \frac{1}{T_{is}} \sum_{t: s(i,t)=s} q_{it}^k - \frac{1}{T_{i,s-1}} \sum_{t: s(i,t)=s-1} q_{it}^k. \quad (4)$$

To express the estimation error compactly, define the average firm-level shock during spell s and its change across adjacent spells as

$$\begin{aligned} \bar{e}_{is}^k &\equiv \frac{1}{T_{is}} \sum_{t: s(i,t)=s} e_{it}^k, \\ \Delta \bar{e}_{is}^k &\equiv \bar{e}_{is}^k - \bar{e}_{i,s-1}^k. \end{aligned} \quad (5)$$

Substituting Equation (1) into Equation (4), the firm fixed effect cancels, and the estimated spell change can be written as

$$\Delta \bar{z}_{is}^k = \Delta z_{is}^k + \Delta \bar{e}_{is}^k. \quad (6)$$

The second term is the estimation error in the measured change in the CEO-spell component.

Under strict exogeneity,

$$E[\Delta \hat{z}_{is}^k | \Delta z_{is}^k] = \Delta z_{is}^k. \quad (7)$$

Thus, under strict exogeneity, the estimated change is unbiased regardless of spell length. Unbiasedness does not, however, make the estimate precise. When spells are short, $\Delta \hat{e}_{is}^k$ need not be close to zero, so $\Delta \hat{z}_{is}^k$ remains noisy. Increasing the number of firms estimates the distribution of these noisy changes more precisely but does not remove the noise within each change. Consequently, the variance of estimated changes and their covariance with other outcomes can differ systematically from the corresponding moments of the true changes.

We study how a firm outcome changes when firm i replaces CEO $s - 1$ with CEO s . Let q_{it}^j denote outcome j of firm i in year t . The outcome j need not be the same variable as the measure k used to construct the CEO-spell change. For example, we may measure the change in the CEO-spell component using revenue and study how employment, capital, profitability, labor productivity, or exporter status changes when the firm changes CEO.

Let g_{is} denote the first year of CEO spell s at firm i . For event time h , define the outcome contrast

$$\Delta_h q_{is}^j = q_{i,g_{is}+h}^j - q_{i,g_{is}-1}^j. \quad (8)$$

Thus, the outcome in each event year is measured relative to the final year of the preceding CEO spell. The same argument applies to more general linear contrasts, including the difference between average post-transition and pre-transition outcomes. This notation keeps separate the variable k used to measure the CEO-spell change and the outcome j whose response we study. They may coincide, as when revenue measures both objects, or differ, as when revenue measures the CEO-spell change and employment is the outcome.

2.2 Second moment biases

The short CEO spells bias both the variance $\text{Var}(\Delta \hat{z}_{is}^k)$ and the covariance between $\Delta \hat{z}_{is}^k$ and the change in the firm attribute, $\text{Cov}(\Delta \hat{z}_{is}^k, \Delta_h q_{is}^j)$.

Bias in the variance. The noise in the estimated CEO-spell change is $\Delta \hat{e}_{is}^k$, the difference between the average firm-level shocks in spells s and $s - 1$.⁵ Let the variance of this estimation error be

$$B^k = \text{Var}(\Delta \hat{e}_{is}^k) \geq 0. \quad (9)$$

Strict exogeneity makes Δz_{is}^k uncorrelated with this estimation error. Therefore,

$$\text{Var}(\Delta \hat{z}_{is}^k) = \text{Var}(\Delta z_{is}^k) + B^k. \quad (10)$$

Equation (10) shows that the observed dispersion combines genuine heterogeneity in CEO-spell changes with noise from firm-level shocks. A variance decomposition based directly on $\Delta \hat{z}_{is}^k$ therefore overstates the variance of the true changes. The size of B^k depends on both spell lengths and the temporal covariance of shocks. Short

⁵For formal derivations of the biases, see Appendix 6.

spells use fewer observations to estimate each average. Correlation within a spell determines how much independent information those observations contain, while correlation across the two spells affects the precision of their difference. Thus, identical spell lengths can generate different biases under different shock processes. Increasing the number of firms estimates the variance of the noisy changes more precisely but does not remove their noise. When spell lengths remain bounded,

$$\widehat{\text{Var}}(\Delta \hat{z}_{is}^k) \xrightarrow{P} \text{Var}(\Delta z_{is}^k) + B^k. \quad (11)$$

The probability limit distinguishes large- N from large- T asymptotics. With many firms but short spells, the sample variance becomes precise around a moment that still contains B^k . The bias disappears only when the number of observations within each spell grows and dependence is weak enough for the spell-average shocks to converge to zero.

Bias in the covariance. Let A_h^{jk} denote the covariance between $\Delta \hat{e}_{is}^k$ and the outcome contrast:

$$A_h^{jk} = \text{Cov}(\Delta \hat{e}_{is}^k, \Delta_h q_{is}^j). \quad (12)$$

The covariance based on the estimated CEO-spell change is therefore

$$\text{Cov}(\Delta \hat{z}_{is}^k, \Delta_h q_{is}^j) = \text{Cov}(\Delta z_{is}^k, \Delta_h q_{is}^j) + A_h^{jk}. \quad (13)$$

Equation (13) shows that A_h^{jk} distorts the covariance because the average-shock difference $\Delta \hat{e}_{is}^k$ may also enter, or be correlated with shocks entering, $\Delta_h q_{is}^j$. When $j = k$, the same outcome shocks can enter both objects directly. When $j \neq k$, the distortion depends on cross-outcome shock covariances. Because the outcome contrast changes with event time, the magnitude and sign of A_h^{jk} may also vary with h . Short spells keep these shocks from averaging out. Unlike classical measurement error, the noise can therefore affect both the variance of the explanatory variable and its covariance with the outcome. This source of bias disappears as spell lengths grow and firm-level shocks average out.

2.3 Implications for commonly reported statistics

Equations (11) and (13) establish the distinction needed below: $\Delta \hat{z}_{is}^k$ is mean-unbiased, but its variance and covariance need not equal the corresponding moments of the true change. We study two functions of these moments. The first is R^2 , which measures the explanatory power of the true CEO-spell change. The second is the projection coefficient β_h^{jk} , which describes its relationship with $\Delta_h q_{is}^j$.⁶ The two statistics answer different economic questions. The R^2 is a unit-free measure of the share of outcome variation linearly associated with the CEO-spell change. The coefficient β_h^{jk} measures the outcome response per unit change in the CEO-spell component and retains the scale of the variables. Because R^2 squares the covariance and also depends on the variance of the outcome contrast,

⁶Instead of event-time projection coefficients, one can also estimate an overall projection coefficient relating Δz_{is}^k to the difference between average post-transition and pre-transition outcomes. Because the bias of this coefficient is analogous to the bias of the event-time coefficients, we do not show it here.

the same A_h^{jk} and B^k can distort the two statistics differently. We therefore derive both rather than treat them as interchangeable measures of CEO importance.

The naive R^2 is

$$\widehat{R}_{h,naive}^{2,jk} = \frac{\widehat{\text{Cov}}(\Delta z_{is}^k, \Delta_h q_{is}^j)^2}{\widehat{\text{Var}}(\Delta z_{is}^k) \widehat{\text{Var}}(\Delta_h q_{is}^j)}. \quad (14)$$

and its probability limit is:

$$\text{plim } \widehat{R}_{h,naive}^{2,jk} = \frac{[\text{Cov}(\Delta z_{is}^k, \Delta_h q_{is}^j) + A_h^{jk}]^2}{[\text{Var}(\Delta z_{is}^k) + B^k] \text{Var}(\Delta_h q_{is}^j)}. \quad (15)$$

This is generally biased because both the covariance in the numerator and the variance of the estimated CEO-spell change in the denominator are contaminated by short-spell estimation error. This expression shows why the direction of the R^2 bias is not mechanical. The nonnegative variance bias B^k enters the denominator, while A_h^{jk} shifts the covariance before it is squared in the numerator. Depending on the sign and magnitude of the true covariance and A_h^{jk} , the two distortions can reinforce or offset one another. Correcting only the variance or only the covariance would therefore combine moments referring to different underlying objects.

We are also interested in the regression coefficients of the outcome contrast on the estimated CEO-spell change at various event times. The population coefficient is

$$\beta_h^{jk} = \frac{\text{Cov}(\Delta z_{is}^k, \Delta_h q_{is}^j)}{\text{Var}(\Delta z_{is}^k)}. \quad (16)$$

This coefficient describes how the change in outcome j at event time h comoves with the change in the CEO-spell component measured using variable k .

Because the true CEO-spell change is unobserved, the natural estimator introduced in Equation 4 replaces Δz_{is}^k with $\Delta \widehat{z}_{is}^k$:

$$\widehat{\beta}_{h,naive}^{jk} = \frac{\widehat{\text{Cov}}(\Delta \widehat{z}_{is}^k, \Delta_h q_{is}^j)}{\widehat{\text{Var}}(\Delta \widehat{z}_{is}^k)}. \quad (17)$$

Combining the variance and covariance decompositions, the probability limit of the naive estimator is

$$\text{plim } \widehat{\beta}_{h,naive}^{jk} = \frac{\beta_h^{jk} \text{Var}(\Delta z_{is}^k) + A_h^{jk}}{\text{Var}(\Delta z_{is}^k) + B^k}. \quad (18)$$

This differs from classical attenuation. Although B^k increases the denominator, A_h^{jk} simultaneously shifts the numerator and may have either sign. The naive coefficient can therefore be above or below β_h^{jk} . In special cases the two distortions may be proportional and cancel in the ratio, but strict exogeneity alone does not imply such an offset.

The covariance distortion is particularly important in event studies because Equation (4) uses outcomes throughout the pre- and post-transition CEO spells. Consider a positive firm-level shock that begins before the transition and persists into the incoming CEO's

spell. The shock raises outcomes used to estimate the incoming spell component, so it enters $\Delta \widehat{z}_{is}^k$. Because it began earlier, it is also correlated with pre-transition outcome contrasts. The estimated CEO-spell change can then appear to predict outcomes before the incoming CEO arrives, even when the true change does not. Persistence is essential to this mechanism: short spells create estimation noise, while serial correlation links that noise to outcomes at other event times. Under the standard assumption of exogenous mobility, the true coefficient satisfies $\beta_h^{jk} = 0$ for $h < 0$. Nevertheless, the probability limit of the naive estimator is

$$\text{plim } \widehat{\beta}_{h,naive}^{jk} = \frac{A_h^{jk}}{\text{Var}(\Delta z_{is}^k) + B^k}, \quad (19)$$

which need not be zero. Serially correlated firm-level shocks can therefore generate apparent pre-trends even when the true CEO-spell change is unrelated to outcomes before the transition.

The same problem can distort the estimated post-transition dynamics. A true relationship that emerges immediately may appear to build gradually, decline over time, or overshoot, depending on how the estimation error in the CEO-spell change covaries with the outcome contrast at different event times. The observed event-study pattern may therefore partly reflect the temporal covariance structure of firm-level shocks rather than the timing of the change associated with the CEO spell.

The variance bias B^k is nonnegative, whereas the covariance distortion A_h^{jk} may have either sign and can vary across event time. Consequently, R^2 , projection coefficients, and event-study dynamics must be constructed from corrected variances and covariances; correcting only one component is insufficient. In particular, short-spell noise can create spurious pre-trends or reshape post-transition dynamics even under strict exogeneity. The next section shows how matched firms without CEO changes identify both bias terms.

3 PLACEBO-CONTROLLED DEBIASING

Our approach constructs placebo CEO transitions to identify and remove the bias terms A_h^{jk} and B^k without specifying the full covariance structure of firm-level shocks.

The key insight of the method is that the short-spell bias depends on two objects. The first is the design matrix, which depicts the construction of the estimated CEO-spell change and the outcome change relative to the reference year, including the pre- and post-spell lengths, transition dates, and event-time comparisons. The second is the variance and covariance structure of firm-level shocks. A placebo sample that reproduces these features but has no true CEO-spell change can therefore identify the bias under an appropriate restriction on the shock process.

Construction of the placebo sample and identification restriction. We match each firm that moves from CEO spell $s-1$ to spell s with a firm that is similar on observables and whose CEO does not change over the full period spanned by the treated firm's two adjacent CEO spells, from the first year of spell $(s-1)$ through the last year of spell (s) . We match firms on characteristics such as one-digit sector, birth cohort in bins of three years and whether maximum employment over the firm's lifetime is below 10 or at least 10. We assign each

matched firm a placebo transition in the same calendar year g_{is} as the corresponding actual transition. This ensures that treated and placebo firms are compared over the same macroeconomic period. We divide placebo firms' uninterrupted CEO spell into artificial pre- and post-transition segments with the same lengths as the two spells in the corresponding treated firm. For notational simplicity, we label these artificial segments $s - 1$ and s .

We then apply identical transformations to the treated and placebo observations. In particular, we use the same pre- and post-spell lengths to construct Δz_{is}^k and the same event-time comparisons to construct $\Delta_h q_{is}^j$.

We use the superscripts "tr" and "pl" to distinguish moments calculated in the treated and placebo samples. Population moments without a sample superscript continue to refer to firms experiencing actual CEO transitions.

Matching does not by itself identify the correction. Its purpose is to reproduce the treated sample's design and make it plausible that treated and placebo firms have comparable shock processes. Our identifying assumption is that, after matching firms and reproducing the treated firms' spell lengths and transition years, the covariance structure of firm-level shocks is the same in the treated and placebo samples up to a scalar multiplier. Formally, for every pair of outcomes j and k ,

$$\text{Cov}^{\text{tr}}(e_{it}^k, e_{ir}^j) = c^{jk} \cdot \text{Cov}^{\text{pl}}(e_{it}^k, e_{ir}^j) \quad \text{for every } t, r, \quad (20)$$

for some constant $c^{jk} > 0$. When $j = k$, we write $c^k \equiv c^{kk}$. We refer to this restriction as the *proportional shock covariance restriction*.

This restriction requires, for each pair of outcomes j and k , the temporal pattern of variances, autocovariances, and cross-outcome covariances to be the same in the treated and placebo groups up to the pair-specific scale factor c^{jk} or c^k . The restriction therefore allows the magnitudes of shock variances and cross-outcome covariances to differ systematically between treated and placebo firms.

The restriction must cover covariances across different outcomes because the variable used to measure the CEO-spell change need not be the same as the outcome whose response is being studied. For example, suppose that the CEO-spell change is measured using revenue, while the outcome of interest is employment. The covariance bias then depends on the covariance between the revenue shocks entering the estimated CEO-spell change and the employment shocks entering the outcome contrast. We therefore require cross-outcome covariances, as well as within-outcome autocovariances, to be proportional between treated and placebo firms. When $j = k$, the restriction reduces to proportionality of the autocovariance structure of a single outcome.

The artificial transition in a placebo firm divides one uninterrupted CEO spell into two segments. There is therefore no true change in the placebo firm's CEO-spell component:

$$\Delta z_{is}^k = 0 \quad \text{in the placebo sample.} \quad (21)$$

Nevertheless, $\Delta \bar{e}_{is}^k$ generally differs from zero because the two artificial segments contain different realizations of firm-level shocks.

The estimated placebo change is therefore

$$\Delta \bar{z}_{is}^k = \Delta \bar{e}_{is}^k \quad \text{in the placebo sample.} \quad (22)$$

Consequently, the variance and covariance calculated in the placebo sample contain only the short-spell bias. Under the proportional shock covariance restriction and the matched placebo construction,

$$\widehat{\text{Cov}}^{\text{pl}}(\Delta \bar{z}_{is}^k, \Delta_h q_{is}^j) \xrightarrow{p} \frac{1}{c^{jk}} A_h^{jk}, \quad (23)$$

and

$$\widehat{\text{Var}}^{\text{pl}}(\Delta \bar{z}_{is}^k) \xrightarrow{p} \frac{1}{c^k} B^k. \quad (24)$$

The treated moments contain both the economic object of interest and the short-spell bias, whereas the corresponding placebo moments contain only the bias. Scaling the placebo moments by c^{jk} and c^k therefore recovers the bias terms that contaminate the treated moments.

The scale factors c^{jk} and c^k capture systematic differences in shock covariances and variances between firms that change CEOs and firms that do not. We estimate c^{jk} and c^k using pre-transition within-spell second moments, before the incoming CEO-spell component can affect the firm's outcomes. In the application, we use a minimum-distance estimator that selects c^{jk} and c^k to align a set of treated and placebo pre-transition moments. Under the proportional shock covariance restriction, this estimator converges to the relative scale of the treated and placebo shock processes.

Debiased moments. Subtracting the scaled placebo covariance from the treated covariance gives

$$\begin{aligned} \widehat{\text{Cov}}^{\text{db}}(\Delta \bar{z}_{is}^k, \Delta_h q_{is}^j) &= \widehat{\text{Cov}}^{\text{tr}}(\Delta \bar{z}_{is}^k, \Delta_h q_{is}^j) \\ &\quad - \tilde{c}^{jk} \widehat{\text{Cov}}^{\text{pl}}(\Delta \bar{z}_{is}^k, \Delta_h q_{is}^j). \end{aligned} \quad (25)$$

Similarly, the debiased variance is

$$\widehat{\text{Var}}^{\text{db}}(\Delta \bar{z}_{is}^k) = \widehat{\text{Var}}^{\text{tr}}(\Delta \bar{z}_{is}^k) - \tilde{c}^k \widehat{\text{Var}}^{\text{pl}}(\Delta \bar{z}_{is}^k). \quad (26)$$

Under strict exogeneity, the proportional shock covariance restriction, the matched-design construction, and consistency of $\tilde{c}^{jk}$ and $\tilde{c}^k$, these estimators converge to the corresponding moments involving the true CEO-spell change:

$$\widehat{\text{Cov}}^{\text{db}}(\Delta \bar{z}_{is}^k, \Delta_h q_{is}^j) \xrightarrow{p} \text{Cov}(\Delta z_{is}^k, \Delta_h q_{is}^j), \quad (27)$$

and

$$\widehat{\text{Var}}^{\text{db}}(\Delta \bar{z}_{is}^k) \xrightarrow{p} \text{Var}(\Delta z_{is}^k). \quad (28)$$

The important feature of these results is that consistency does not require CEO spells to become long. The number of observations within each spell may remain fixed while the number of observed treated and placebo spell pairs grows. The placebo sample identifies the nonvanishing short-spell bias that would otherwise remain in the probability limits of the treated moments.

For regression coefficients and other nonlinear functions of second moments, we first debias the underlying moments and then

form the relevant statistic. For example, the debiased projection coefficient is

$$\widehat{\beta}_{h,db}^{jk} = \frac{\widehat{\text{Cov}}^{db}(\Delta \tilde{z}_{is}^k, \Delta h_{is}^j)}{\widehat{\text{Var}}^{db}(\Delta \tilde{z}_{is}^k)}. \quad (29)$$

Thus, the correction is performed at the level of the covariance and variance. In general, it is not equivalent to subtracting a placebo regression coefficient from the treated regression coefficient. Correlations, R^2 statistics, and other functions of second moments are constructed in the same way: we first correct each underlying variance and covariance and then apply the nonlinear transformation.

The placebo-control approach has three advantages. First, it does not require a parametric model for the full covariance structure of firm-level shocks. Second, by reproducing the treated sample's spell lengths, transition dates and event-time comparisons, it accommodates short and heterogeneous CEO spells and unbalanced panels. Third, it applies separately at each event time, allowing us to correct both spurious pre-trends and distorted post-transition dynamics.

4 MONTE CARLO DESIGN AND INTERPRETATION

This section asks two questions. First, does the placebo correction recover the variance, covariance, and event-time response when its identifying assumptions hold in a short and persistent panel? Second, what remains in the estimates when those assumptions fail? We answer these questions in four steps. We first describe the simulated firms and CEO spells. We then specify the firm-level shock process and the assignment of the incoming CEO-spell component. Next, we identify the parameter restrictions that correspond to the assumptions in Section 3. Finally, we change these parameters one at a time and compare the naive and placebo-controlled estimates. Figure 1 gives the main results as complete event-time paths; Table 1 reports the corresponding moments.

We begin with 50,000 candidate CEO transitions. For each transition, we draw continuous waiting times independently from an exponential distribution with rate 0.2 and round them up to obtain the integer-valued lengths T_{1i} and T_{2i} of the first and second CEO spells. The resulting discrete spell lengths are geometric, with an annual probability of CEO change equal to $1 - \exp(-0.2)$. In the main experiments, we retain transitions for which both spells last at most five years. We then create an actual firm and a placebo firm with the same spell lengths and transition year. The actual firm receives a new CEO-spell component at the transition. The placebo firm does not, but we divide its observations into artificial pre- and post-transition spells at the same event date.

The identification results above do not require firm-level shocks to follow an AR(1) process. A simulation, however, must specify how shocks evolve over time. We use an AR(1) process because persistent shocks make observations within a short CEO spell highly dependent and therefore make the small-sample problem consequential. For group $g \in \{\text{tr}, \text{pl}\}$, the shock is

$$e_{it}^g = \rho_g e_{i,t-1}^g + \varepsilon_{it}^g, \quad \varepsilon_{it}^g \sim N(0, \sigma_{\varepsilon,g}^2), \quad (30)$$

where tr denotes actual transitions and pl denotes placebo transitions. We allow persistence ρ_g and innovation variance $\sigma_{\varepsilon,g}^2$ to differ

across the two groups. In the baseline, shocks are highly persistent in both groups: $\rho_{\text{tr}} = \rho_{\text{pl}} = 0.9$. High persistence is therefore part of the environment that the method is intended to handle, not itself a violation of identification.

We normalize the firm fixed effect and the first CEO-spell component to zero. Let $D_i = 1$ for an actual transition and $D_i = 0$ for a placebo transition, and let t_i denote the first year of the second spell. The simulated outcome is

$$\ln R_{it} = e_{it}^g + D_i \mathbf{1}\{t \geq t_i\} z_i. \quad (31)$$

Before the transition, outcomes contain only the firm-level shock. From the transition year onward, the actual firm also receives the permanent CEO-spell component z_i . The placebo firm never receives this component. The population event-time coefficient is therefore zero before the transition and one afterward when the incoming component is assigned independently of the shock process.

The final part of the data-generating process determines whether that independence holds. We draw

$$z_i = \gamma \ln R_{i,t_i}^{\text{pre}} + v_i, \quad v_i \sim N(0, \sigma_z^2), \quad (32)$$

where $\ln R_{i,t_i}^{\text{pre}}$ is the transition-year outcome before adding z_i , and $\sigma_z = 1$. When $\gamma = 0$, the incoming CEO-spell component is independent of the firm's current and past shocks. When $\gamma > 0$, firms experiencing better performance at the transition receive systematically different CEO-spell components. Because the AR(1) process carries past shocks into the transition-year outcome, this specification allows the CEO change to respond to the firm's prior performance.

The special case covered by our identification argument sets

$$\gamma = 0 \quad \text{and} \quad \rho_{\text{tr}} = \rho_{\text{pl}}. \quad (33)$$

The first equality imposes strict exogeneity. The second gives treated and placebo shocks the same temporal covariance pattern. Their innovation variances may still differ: when the autoregressive coefficients are equal, a difference in $\sigma_{\varepsilon,g}^2$ scales the entire covariance matrix by a constant and satisfies the proportional shock covariance restriction. The baseline imposes the still simpler case $\sigma_{\varepsilon,\text{tr}} = \sigma_{\varepsilon,\text{pl}} = \sqrt{0.5}$. It therefore evaluates the placebo correction in a setting where both identifying conditions hold, despite short spells and common persistence of 0.9.

The remaining experiments change one part of this special case at a time. Panel B sets $\rho_{\text{tr}} = 0.9$ and $\rho_{\text{pl}} = 0.8$. The shock covariance matrices then differ in shape, not only scale, so the proportional covariance restriction fails. Panels C and D keep persistence equal but raise the actual firms' innovation standard deviation to $1.2\sqrt{0.5}$. This difference is proportional: Panel C omits the scale adjustment, while Panel D applies it to the same simulated observations. Panel E combines excess variance with unequal persistence, setting $\rho_{\text{tr}} = 0.8$ and $\rho_{\text{pl}} = 0.9$. Panel F instead keeps the baseline shock process and sets $\gamma = 0.1$, so that the incoming CEO-spell component responds to transition-year performance and strict exogeneity fails.

Figure 1 provides the clearest summary of the experiments. The red line reports the naive OLS coefficient, and the blue line reports the ratio of the placebo-debiased covariance to the placebo-debiased variance. Event time $t = -1$ is the reference year. Under the maintained assumptions, the target is zero before the CEO transition and one in every post-transition year.

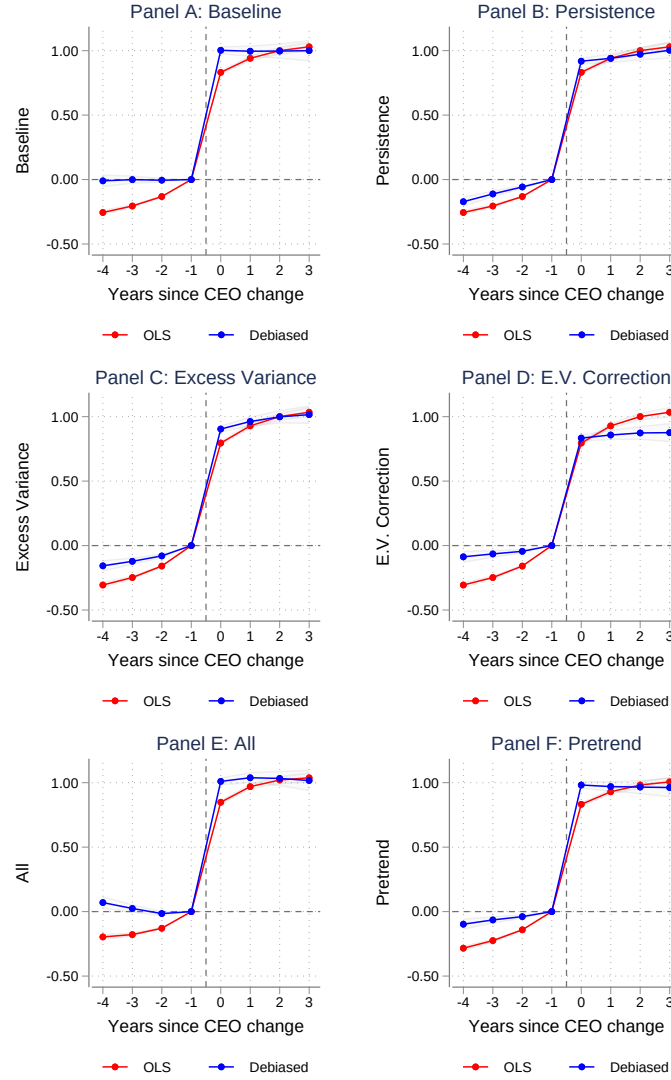


Figure 1: Monte Carlo Event Studies Under Six Scenarios

Notes: Event-time coefficients from the naive OLS estimator (red) and the placebo-controlled estimator (blue). Shaded areas report 95% confidence intervals. Event time $t = 0$ is the first year of the second CEO spell, and $t = -1$ is the reference year. Panel A uses the baseline parameters $\rho_1 = \rho_0 = 0.9$ and $\sigma_{\varepsilon,1} = \sigma_{\varepsilon,0} = \sqrt{0.5}$. Panel B sets $\rho_1 = 0.9$ and $\rho_0 = 0.8$. Panels C and D set $\sigma_{\varepsilon,1} = 1.2\sqrt{0.5}$; Panel D uses the same simulated data as Panel C but applies the excess-variance correction. Panel E combines excess variance with $\rho_1 = 0.8$ and $\rho_0 = 0.9$. Panel F sets $\gamma = 0.1$, making the incoming CEO-spell component depend on the transition-year outcome.

Panel A is the central result. The naive estimates show a strong negative pretrend and a gradual post-transition response even though the population response is exactly zero before the transition and jumps immediately to one afterward. The pretrend is entirely spurious: the same short-spell shocks enter both the estimated CEO-spell change and the event-time outcome. Once the placebo moments are removed, the blue coefficients are approximately zero before the transition and one afterward. Thus, when the identifying assumptions hold, the method works essentially perfectly. More importantly, the panel shows that even a visually

strong pretrend need not signal anticipation or endogenous treatment timing; it can be generated by short-panel estimation noise alone.

The remaining panels ask what happens away from this benchmark. Panel B gives actual and placebo shocks different persistence. This violates the proportional covariance restriction, and the correction no longer removes the entire pretrend or fully recovers the impact response. It nevertheless moves both parts of the profile toward their population targets. The failure is therefore informative rather than catastrophic: violations of the covariance restriction

make debiasing less accurate, but the adjustment still removes part of the short-panel distortion.

Panels C and D isolate excess variance. They use the same simulated observations, in which actual firms have more volatile innovations than placebo firms. Without the scale adjustment, Panel C retains a negative pretrend and understates the impact response. Applying the excess-variance correction in Panel D removes most of the pre-transition distortion and brings the post-transition path much closer to the benchmark. This is the second main result. A difference in shock variance is an empirically important departure from the baseline, but it is one that the explicitly scaled placebo correction handles well.

Panel E combines excess variance with unequal persistence. The placebo-controlled estimates again move substantially toward zero before the transition and toward one afterward, although small pre-transition deviations and a post-transition overshoot remain. Across Panels B and E, violations of the identifying restrictions make the method less precise, but debiasing continues to move the estimates in the right direction.

Panel F serves as an informal power test: it asks whether the method mechanically erases any pretrend, including one generated by the treatment itself. Here the incoming CEO-spell component depends on performance at the transition, so the pretrend reflects genuine endogenous assignment rather than only short-spell noise. The correction does not eliminate it. This distinction is built into the estimator. It removes the pretrend found in the placebo group, scaled when necessary for excess variance; it does not subtract a pretrend that appears only in the actual treatment process.

The figure therefore delivers three conclusions. Under the maintained assumptions, the correction removes a strong spurious pretrend and recovers the sharp population response almost exactly.

When the covariance restrictions fail, the correction is incomplete but still helpful. Excess variance can be handled particularly well by scaling the placebo moments, while a true treatment-related pretrend remains visible.

Table 1 gives a numerical representation of these findings rather than a separate qualitative lesson. It reports the variance of the estimated CEO-spell change, its covariance with the outcome change, and the resulting R^2 , both over the full post-transition spell and at event time $t = 3$.

The magnitudes show that the biases are quantitatively important. In the baseline, the naive variance and covariance are 1.776 and 1.764 even though both population moments equal one. The placebo correction recovers 1.000 and 1.001. Correcting these moments lowers the full-spell R^2 from 0.889 to 0.508 and the event-time R^2 from 0.779 to 0.382. A naive event-study coefficient near one can therefore coexist with large errors in the underlying variance decomposition and with the strong spurious pretrend in Panel A.

The excess-variance columns quantify the comparison between Panels C and D. Applying the scale correction to the same simulated data lowers the corrected covariance from 1.334 to 1.174, the full-spell R^2 from 0.555 to 0.430, and the event-time R^2 from 0.432 to 0.322. The adjustment does not force every component moment to one, but it substantially reduces the economically relevant distortion. The long-panel experiment reaches the same broader conclusion: even when the maximum spell length rises from 5 to 20 years, the naive variance and covariance remain 2.029 and 1.821, whereas the placebo correction recovers 1.015 and 1.013. Short-panel bias is therefore large enough to matter for substantive conclusions, and it does not disappear merely because some CEO spells are longer.

Table 1: Monte Carlo Estimates by Scenario

	Baseline	Excess Variance	E. V. with Correction	Long Panel
$\hat{Var}(dz)$ (OLS)	1.776	2.114	2.114	2.029
$\hat{Var}(dz)$ (debiased)	1.000	1.339	1.339	1.015
$\hat{Cov}(dz, dy)$ (OLS)	1.764	2.096	2.096	1.821
$\hat{Cov}(dz, dy)$ (debiased)	1.001	1.334	1.174	1.013
R^2 (OLS)	0.889	0.868	0.868	0.766
R^2 (debiased)	0.508	0.555	0.430	0.474
R^2 at $t = 3$ (OLS)	0.779	0.747	0.747	0.663
R^2 at $t = 3$ (debiased)	0.382	0.432	0.322	0.374

Notes: The simulation begins with 50,000 candidate CEO transitions. Continuous waiting times are independent exponential draws with rate 0.2 and are rounded up to obtain geometric spell lengths; the implied annual probability of CEO change is $1 - \exp(-0.2)$. Unless noted otherwise, transitions are retained when both spells last at most five years, $\sigma_z = 1$, $\sigma_{\varepsilon,1} = \sigma_{\varepsilon,0} = \sqrt{0.5}$, and $\rho_1 = \rho_0 = 0.9$. In the excess-variance columns, $\sigma_{\varepsilon,1} = 1.2\sqrt{0.5}$. The correction column uses the same simulated data as the preceding column but applies the excess-variance adjustment. The long-panel column raises the maximum spell length to 20 years. OLS denotes the naive estimate; debiased denotes the placebo-controlled estimate. The full-spell R^2 uses the ATET specification based on all retained years. The event-time R^2 uses the outcome at $t = 3$ relative to the final pre-transition year.

5 APPLICATION: CEO REPLACEMENTS IN HUNGARIAN FIRMS

We apply the placebo-controlled debiasing method to CEO replacements in the universe of Hungarian firms. The Hungarian data offer an ideal setting, as they encompass the entire population of businesses over three decades, unlike most other studies that focus exclusively on public firms or large corporations [3, 6, 20]. In this section, we use our method to obtain estimates of changes in firm behavior around CEO replacements. In the following, we describe our data and present results on several firm outcomes for two measures of CEO change (Δz): growth (measured by log revenue) and profitability (measured by the return on assets). The outcomes are measures of the output (revenue and an exporter dummy), inputs (capital and labor), and efficiency (return on assets and labor productivity).

5.1 Data Construction and Descriptive Statistics

Our analysis combines two administrative datasets. The Firm Registry is maintained by Hungarian Corporate Courts and contains records on all individuals authorized to act on behalf of firms. These records include CEOs and other executives with signatory rights, tracked through a temporal database where each entry reflects representation status over specific time intervals. Updates occur when positions change, personal identifiers are modified, or reporting standards evolve. The registry provides names, addresses, dates of birth (from 2010), and mother’s names (from 1999), though numerical identifiers exist only from 2013 onward. Information on CEOs is linked with tax identifiers to the Balance Sheet Database (maintained by the National Tax and Customs Administration), which has annual balance sheets and income statements, as well as employment, industry code and ownership (state, domestic, foreign) for all Hungarian firms with double-entry book-keeping. The two datasets cover 1,063,172 firms for the period between 1992 and 2024, yielding 9,627,484 firm-year observations before sample restrictions.

Finding CEOs among all individuals in the Firm Registry requires heuristics since job titles are inconsistently recorded. When explicit “managing director” titles are recorded, we use them directly. For remaining cases, we assume all representatives are CEOs if three or fewer exist at the firm. When more than three representatives are present, we assign CEO status based on continuity with previously identified CEOs. Time spans between appointments are often unclosed or non-contiguous. In these cases we use information from adjacent years to identify the CEO. In all other cases we do not identify a CEO in that firm-year (ADD PROPORTION). These procedures help identifying CEOs for XX% of all firm-years.

Constructing CEO time series across firms poses several challenges.⁷ Before 2013, no numerical identifiers existed, requiring entity resolution based on names, addresses, mother’s names, and birthdates. We link records across these dimensions to create unique person identifiers, enabling tracking across firms and over time. Matching quality improves after 1999 (when mother’s names reporting begins) and 2010 (when birthdates reporting starts), but the 1990s data also achieve reasonable coverage through name and address matching.

After the construction of the CEO time series, we take a yearly snapshot of the data because firm level information is available yearly. We consider a person to be the CEO of a given firm in a year if she was recorded as the CEO on June 21.

Firms with multiple CEO changes are handled by splitting their history into adjacent transition pairs. After collapsing firm-CEO spells, intermediate spells are duplicated so that a firm with spells 1/2/3 contributes two transitions, $1 \rightarrow 2$ and $2 \rightarrow 3$. Each transition defines a pseudo-firm, indexed by a constructed identifier that groups the true firm with the relevant spell window. Fixed effects are estimated separately for these pseudo-firms, so outcomes around different transitions are allowed to have distinct firm intercepts. These pseudo-firms are not treated as independent statistical units. We retain the original firm identifier and cluster standard errors by the true firm.

We apply several restrictions to create a sample suitable for the analysis. We drop firm-years without a CEO and with more than 12 CEO spells during the observed period to reduce noise from misclassified transitions. We drop firms ever having more than 2 simultaneous CEOs to avoid complex governance structures that complicate identification. We drop the finance sector due to specialized accounting frameworks and regulatory environments. We drop the first year of firms to avoid mixing stock variables with flow variables observed over less than 12 months. Finally, we drop firms which never reach 3 employees during the observation period. This filter removes a substantial portion of observations but eliminates shell companies, tax optimization vehicles, and self-employment arrangements masquerading as corporations. The remaining firms represent genuine businesses with meaningful economic activity where management quality affects performance. For each CEO replacement, we keep a maximum of 4 precedent and subsequent firm-years (the latter includes the replacement year), forming a symmetric interval around the CEO change.

⁷In this analysis, we use the time series only to identify CEO replacements within firm, and do not use CEO transitions from one firm to the other.

The final sample has 471 thousand firms and 4,363,067 firm-years. Appendix Table B1 shows the number of firms for several years along with the number of firms and their average employment each year. The number of Hungarian firms increased abruptly during the nineties, from 52 thousand in 1992 to 168 thousand in 2000. The number of firms was quite stable during the next 13 years, and it grew again as the country went out of the great recession of 2008. In 2023 (the last year of our data), the number of firms was over 250 thousand. The average employment size was rather large in early years (19) as it was strongly influenced by the large socialist companies. But employment size quickly declined as small firms entered the data: by 1997 it fell to 9 and remained between 7 and 9 throughout of the subsequent period.

We observe 772 thousand distinct CEOs in the data. They typically manage the firm alone, but 158 thousand (20.4%) have a period when two CEOs serve together. This affected about 128/855 thousand firms (firm-years). Thus, multiple CEOs are not a rare event in our data, as 27% of firms and 19% of firm-years have two CEOs.

In our data, 48% of firms replace their CEO. Among these firms, the typical company experiences one CEO change, as the median of the distribution is 1. Multiple CEO replacements over the 23 year long period covered by our data are uncommon; even at the 75th percentile, firms have experienced only 2 CEO changes. The proportion of firms with two CEOs serving at the same time is 28%. The average (median) years we observe the firm is XX (10), but we have firms which survive a long time (the 75 percentile of firm age distribution is 19).

The table also shows the number of firms with a CEO replacement. Depending on the year, we observe between 3,900 and 11,000 CEO switches, corresponding to a proportion of firms with a CEO replacement of approximately 5% to 9%. In total, we have 275 CEO switches, which provides a solid basis for estimating the effect of CEO changes.

Appendix Table B2 shows the industrial distribution of firms. The largest sectors are Wholesale-Retail with 27% of firms, followed by business services (17%). Industrial and construction sectors each have a share around 11-13% and agriculture only 3%. The remaining 28% of firms belong to various service sectors.

Firms which experience CEO transitions are different from those which have only one CEO during their whole observed life span. As Table 2 shows, firms which switch CEOs, have more employees (9.6 vs. 6.2), have their output almost twice as large, and their labor productivity is also 12% higher. Return on assets, on the contrary, is smaller by 14%. The table also shows the changes of these variables around a CEO transition (measured as the proportional change of the average value of the variable in the subsequent and precedent 4 years). Employment declines by 11% while sales grow by 6%, leading to an increase in labor productivity by 13%. ROA declines by 3%.

The tenure of the first CEO of the firm is shown in Figure 2.⁸ Short tenures abound: 18% of CEOs have a one-year long tenure, and 12% two-year long tenure in a given year. To assess the prevalence of the CEOs by the maximum tenure, we weight tenure by the number of years serving. This statistic weights up CEOs who serve for a long time. The distribution of this statistic is shown in the

bottom panel of Figure 2: the typical CEO serves between 4 and 10 years. In this range each tenure has a share of 5-6% of all firm-years. The prevalence of longer tenures than 10 years gradually decreases up to 20 years, when it levels again.

5.2 Analysis of Variance and Event-Time Estimation

Table 3 presents the naive and placebo-controlled debiased R^2 for various firm outcomes: revenue, inputs (labor, capital) and performance (return on assets and labor productivity). The debiased R^2 s decline substantially relative to the simple OLS estimates. The average value of the R^2 from the naive estimation is 0.658 while our debiased method decreases them to 0.263, or to 40% of the naive R^2 s.

While the debiased R^2 s are indeed much smaller than the naive ones, they still underline the importance of CEOs in our SME sample. They are in the range of 0.25–0.30 (except for ROA when it is 0.19), showing that CEO change alters firm outcomes in an economically meaningful way.

Figure 4 presents the event-time estimations (the reference year is one year before the CEO transition). The coefficients can be interpreted as elasticities (except for ROA, which is not logged) between the firm outcome and δz in various event-years relative to the year precedent to CEO change. The red lines represent the coefficients from the naive estimation and the blue lines from the placebo-debiased estimation. As the Monte-Carlo estimation predicted, pre-trends largely diminish in the observational data when the placebo-controlled debiasing method is used while they are rather large when OLS is used for estimation. Pre-trends are flat and small in magnitude for each firm outcome, and in two cases they are also statistically insignificant. On the contrary, the OLS regression provides increasing pre-trends, which are significant both in statistical and economic terms. To take an example, 4 years before the CEO change labor productivity is 23% lower than one year before the CEO change and presents a constant increase. The placebo controlled debiasing method almost completely removes this spurious pre-trend: the coefficient associated with event time = -4 equals -6%, or about 4 times lower.

Thus, simple OLS regressions would suggest that better CEOs – relative to the their predecessor – go to firms which are at an increasing trajectory in output, inputs and firm performance, while the placebo-controlled debiasing method does not find any evidence of the correlation between CEO value and firm behavior before the CEO change. Estimating unbiased pre-trends is crucial for any empirical analysis, as the reliability of the difference-in-difference estimation is predominantly judged upon the existence of pre-trends. Our exercise demonstrates that the limited mobility bias may show up in ways which may make researchers reject well specified estimations.

The post-treatment coefficients demonstrate the importance of the value of the CEO for all our firm outcomes. The elasticities quickly jump to a high level and are rather stable in the 3 fully treated years.⁹ Their magnitude is between 0.62 for labor productivity and 0.75 for employment (we discuss ROA in the next paragraph).

⁸In this figure, we measure tenure by the maximum tenure of the CEO. The tenure of the second, third, etc. CEO is somewhat shorter, as Appendix Table B3 demonstrates.

⁹Event year 0 is only partly affected by the arrival of the new CEO, as the CEO change may take place any time between June of the year and June of the precedent year.

Table 2: Characteristics of Firms With and Without a CEO Replacement

	lnR	Exporter	lnL	lnK	ROA	lnRL	N
Without CEO Switch	10.0 (1.7)	0.1 (0.3)	1.4 (0.9)	7.9 (2.2)	2.5 (9.8)	8.6 (1.4)	220136
With CEO Switch	10.2 (1.9)	0.1 (0.3)	1.5 (1.0)	8.3 (2.4)	2.2 (9.0)	8.7 (1.4)	231710
Growth	-0.008 (0.185)	-0.204 (0.760)	-0.079 (0.657)	-0.003 (0.284)	-0.057 (374.204)	0.004 (0.192)	
Pre-trend	0.004 (0.114)	-0.154 (0.498)	-0.007 (0.433)	-0.003 (0.240)	-0.080 (85.598)	0.006 (0.125)	

Notes: the first two lines of the table present the mean (standard deviation) firm characteristics with and without CEO replacement. The third and fourth lines show the proportional change of firm characteristics around and before CEO replacement. Around CEO replacement = $(\bar{y}_{after} - \bar{y}_{before}) / \bar{y}_{before}$, before CEO replacement = $(y_{-1} - y_4) / y_{-4}$.

Table 3: Analysis of Variance – Naive and Debiased Estimates

Estimate	lnR	lnL	lnK	ROA	lnRL
R^2 (OLS)	0.677	0.672	0.653	0.642	0.646
R^2 (debiased)	0.309	0.256	0.268	0.185	0.292

Notes: N = 2,551,515. The table presents the naive and placebo-controlled debiased estimated coefficients and the corresponding R^2 of CEO replacement. The variable used for estimating Δz is log revenue (top panel), return on assets (bottom panel). lnR = log revenue; lnL = log labor; lnK = log tangible assets; ROA = return on assets; lnRL = log labor productivity.

For example, a CEO who is 10% more able to raise employment than the CEO preceding her in the position, can actually raise employment by 7.5% after taking into account the noise in the data.

The estimated coefficients of ROA are semielasticities, as we could not log the variable due to negative values. To transform it to an elasticity, we divide the estimated coefficients with the average value of ROA of the firms which experienced a CEO switch, which equal 2.2. This elasticity equals 0.23-0.27, depending on the event year. Thus, ROA is less sensitive to the CEO value than other firm attributes.

The differences between the naive and corrected event-time estimations are smaller for the period subsequent to the CEO transition than for pre-trends, but they are present nonetheless. For output, labor and capital the two estimations provide very similar coefficients but for the two performance measures the debiased estimates are smaller than the naive ones. This difference is about 20 percentage points for ROA and 15% for labor productivity in the last event time studied.

The comparison of the naive and debiased dynamics demonstrate that there is nothing mechanical in our method in the sense that the difference between naive and debiased estimates vary depending on the outcome variable. The pre-trend disappears completely for two variables but only diminishes for the other three. In the post-treatment period the estimated coefficients are sometimes almost identical but there are cases when they are substantially different.

To summarize, the estimations based on our observational data underline the lessons learned from the Monte-Carlo estimations. The placebo-controlled debiasing method demonstrates that the explanatory power of Δz for changes in firm attributes is much lower than OLS regression predict, and the pre-trends in event-time

estimations are spurious in the naive estimates which may lead to the rejection of well-identified estimates.

Do the observable characteristics of CEOs determine firm outcomes? To study this, we split the sample along CEO switches when the incumbent and the new CEO are similar along some dimension or not, and compute the R^2 of the regression. To save space, we do this only for output. The results are in 4. First we split the data based on whether one CEO replaced the other, as opposed to one CEO being continuously present with the company and another joining or leaving the firm (see Section 5.1). The R^2 for the earlier sample is 0.35 and for the latter 0.29. Thus, when there is continuity in the leadership of the firm, CEO change explains less of the variance of output change by 17%. Next, we look into changes between founders and nonfounders and between two nonfounders.¹⁰ Again, we find that when the founder leaves the firms, the R^2 is 17% higher. We also look at gender switch and find that when the incumbent and the new CEO have the same gender, the estimated R^2 is 12% lower.

¹⁰Founders are defined as the first CEO of the firm, while nonfounders are all the other CEOs.

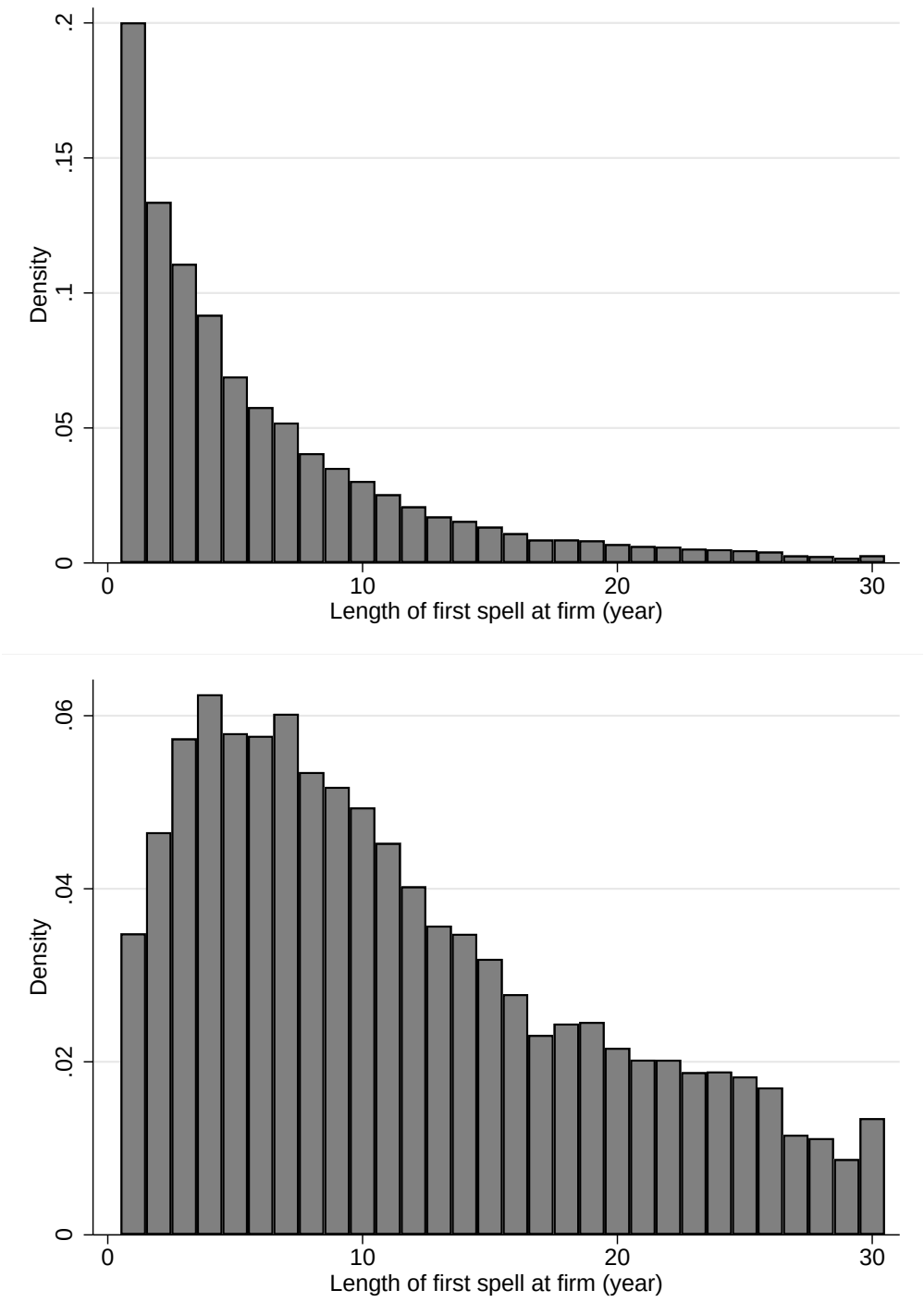


Figure 2: CEO Tenure

Notes: Maximum tenure of the first CEO. Top panel: unweighted; Bottom panel: weighted by the number of years.

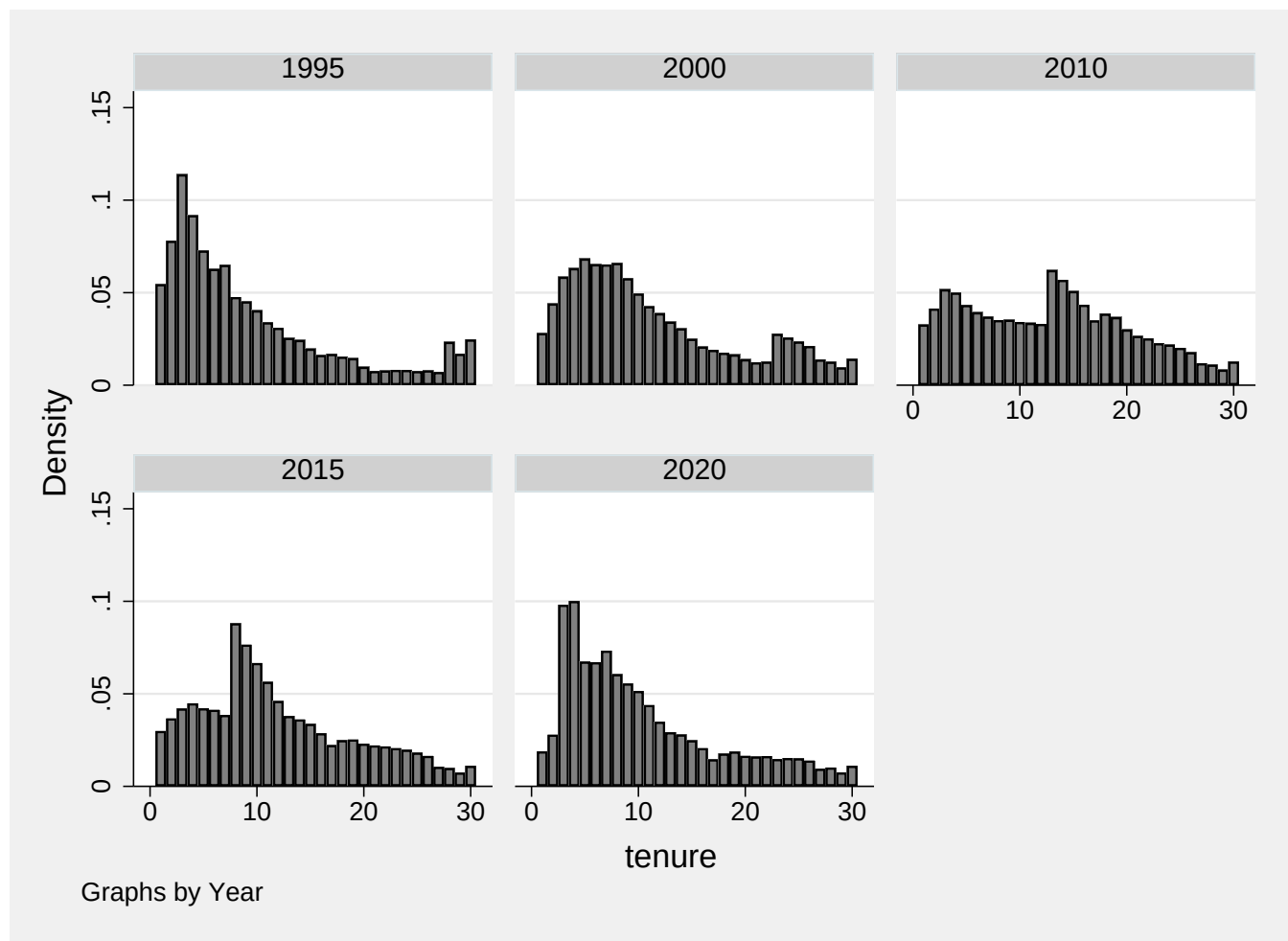


Figure 3: CEO Tenure by Year

Notes: Actual CEO tenure in various years.

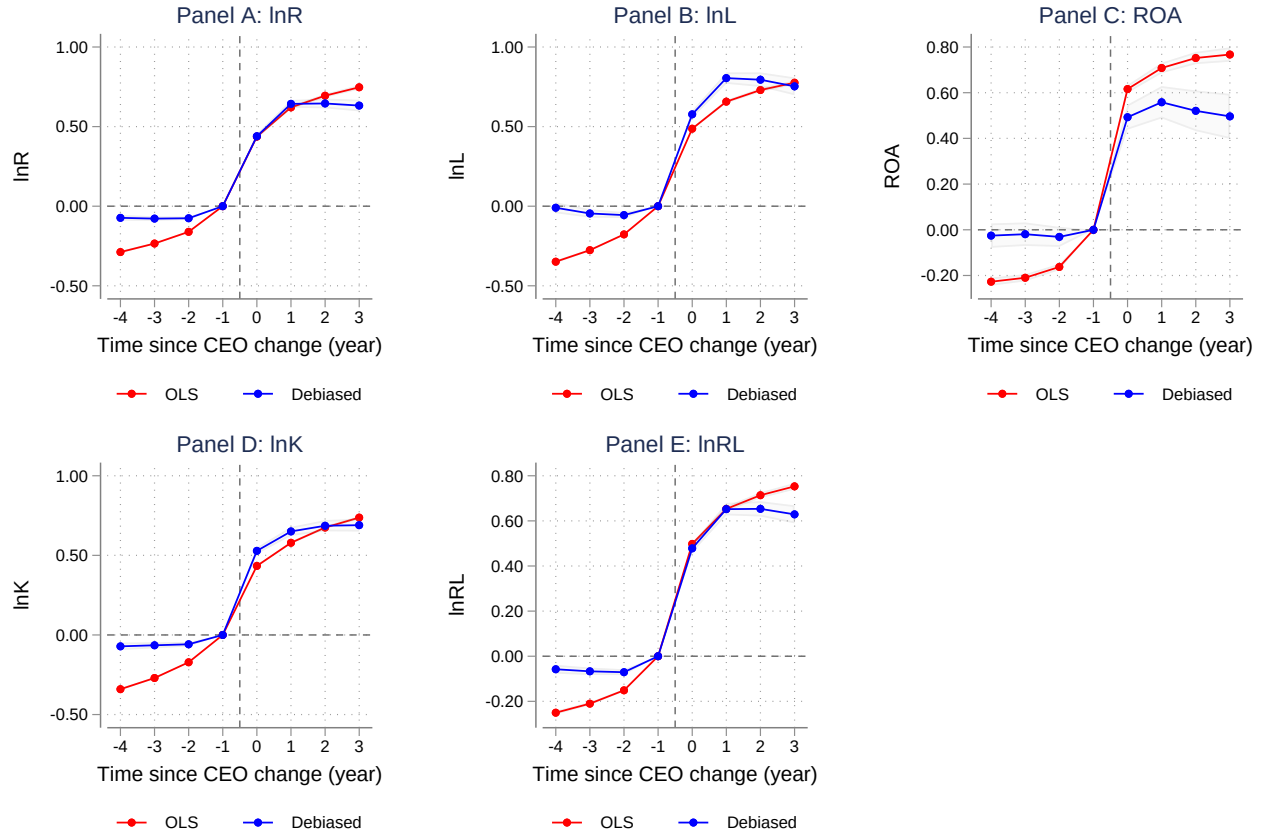


Figure 4: Naive and Debiased Event-Time Estimates Around CEO Transitions

Notes: Event-time dynamics of naive (red line) and placebo-controlled debiased (blue line) coefficients and 95% confidence intervals. Δz measured by $\ln R$. $\ln R$ = log revenue; $\ln L$ = log labor; $\ln K$ = log tangible assets; ROA = return on assets; $\ln RL$ = log labor productivity.

Table 4: Analysis of Variance in Subsamples

Samples	lnR	lnL	lnK	ROA	lnRL	N
One-to-One	0.346	0.284	0.295	0.197	0.330	1909382
Twos	0.277	0.241	0.236	0.188	0.246	2004282
Founder-to-Non	0.390	0.304	0.328	0.224	0.382	1162079
Non-to-Non	0.328	0.295	0.307	0.205	0.315	1631531
Gender switch	0.334	0.269	0.277	0.194	0.308	1913895
No gender switch	0.297	0.254	0.269	0.198	0.283	2002551
Age gap	0.349	0.288	0.303	0.198	0.332	1439578
No age gap	0.335	0.276	0.270	0.194	0.317	1047934
Max Emp. <10	0.311	0.245	0.270	0.168	0.298	1260772
Max Emp. 11-50	0.315	0.275	0.263	0.200	0.269	1003888
Max Emp. 51-100	0.308	0.261	0.245	0.191	0.205	325426
Max Emp. 100+	0.348	0.338	0.278	0.284	0.250	298148
Pre 2000 switch	0.262	0.240	0.247	0.147	0.223	521375
Post 2000 switch	0.330	0.266	0.281	0.207	0.314	2618528

The table presents the placebo-controlled debiased estimated R^2 of CEO replacement for various subsamples defined by CEO switch characteristics. $\ln R$ = log revenue.

6 CONCLUSION

ÁTIRNI We introduce a placebo-controlled difference-in-differences methodology that corrects for small-sample bias in CEO fixed effects estimation. The method constructs matched “placebo” transitions in firms without CEO changes, assigning fictitious transition dates aligned with the actual treatment group. This facilitates the recovery of bias in second moments without the need to model the full error process. Monte Carlo simulations confirm that the approach robustly eliminates bias under persistent shocks, short tenures, and unbalanced panels. It also eliminates spurious pre-trends that arise when error terms are autocorrelated.

Empirical analysis of 60,000 CEO transitions in Hungarian firms reveals that, after correction, CEO effects account for 20–30% of revenue variance, about half of what naive estimates imply. CEO replacements have immediate and persistent effects on firm growth, with no spurious pre-trends.

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APPENDIX A: DERIVATION OF BIAS TERMS

This appendix provides the formal matrix algebra derivation of the bias terms in second moments of estimated CEO effects.

6.1 Matrix Notation Setup

Model setup. Let firm i be observed for T periods with outcome path $\mathbf{y}_i \in \mathbb{R}^T$ that decomposes into a (piecewise-constant) manager-effect path $\mathbf{z}_i$ and shocks $\mathbf{e}_i$. The model is

$$\mathbf{y}_i = \alpha_i + \mathbf{z}_i + \mathbf{e}_i \quad (\text{A1})$$

with

$$\begin{aligned} \mathbb{E}[\mathbf{z}_i \mathbf{z}_i'] &= \Lambda_i = \begin{bmatrix} \lambda_{i11} \otimes \mathbf{1}\mathbf{1}' & \lambda_{i12} \otimes \mathbf{1}\mathbf{1}' \\ \lambda_{i12} \otimes \mathbf{1}\mathbf{1}' & \lambda_{i22} \otimes \mathbf{1}\mathbf{1}' \end{bmatrix}, \\ \mathbb{E}[\mathbf{z}_i \mathbf{e}_i'] &= 0, \quad \mathbb{E}[\mathbf{e}_i \mathbf{e}_i'] = \sigma_i \Sigma \end{aligned} \quad (\text{A2})$$

for changing firms, and

$$\mathbb{E}[\mathbf{z}_i \mathbf{z}_i'] = \Lambda_i = \lambda_{i11} \otimes \mathbf{1}\mathbf{1}', \quad \mathbb{E}[\mathbf{z}_i \mathbf{e}_i'] = 0, \quad \mathbb{E}[\mathbf{e}_i \mathbf{e}_i'] = \sigma_0 \Sigma \quad (\text{A3})$$

for non-changing firms. The first equation is notation for the variance and covariance of manager fixed effects at firm i . The second equation states strict exogeneity (random mobility): every period's shock is mean independent of the entire CEO path. The third assumption allows for arbitrary autocorrelation in shocks over time but assumes that this autocorrelation is the same for changing and non-changing firms, up to a scalar multiplier σ_1/σ_0 (proportional autocovariance).

Projection Matrix and LSDV Estimator. Define design matrix $\mathbf{D}$ and spell-length matrix $\mathbf{T}$. For $T_1 = 2, T_2 = 3$:

$$\mathbf{D} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{bmatrix},$$

and the diagonal matrix of spell lengths $\mathbf{T}$ as

$$\mathbf{T} = \text{diag}(T_1, T_2) = \text{diag}(2, 3).$$

The projection matrix that converts outcomes to within-spell means is

$$\mathbf{P} = \mathbf{D} \mathbf{T}^{-1} \mathbf{D}'.$$

In the example,

$$\mathbf{P} = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 1/3 & 1/3 & 1/3 \\ 0 & 0 & 1/3 & 1/3 & 1/3 \end{bmatrix},$$

$$\Rightarrow \mathbf{P} \mathbf{y}_i = \alpha_i + \begin{pmatrix} \frac{1}{2} \sum_{t=-2}^{-1} (z_{it} + e_{it}) \\ \frac{1}{2} \sum_{t=-2}^{-1} (z_{it} + e_{it}) \\ \frac{1}{3} \sum_{t=0}^{+2} (z_{it} + e_{it}) \\ \frac{1}{3} \sum_{t=0}^{+2} (z_{it} + e_{it}) \\ \frac{1}{3} \sum_{t=0}^{+2} (z_{it} + e_{it}) \end{pmatrix}.$$

Here $\mathbf{P}$ is idempotent, symmetric, and $\mathbf{P} \mathbf{z}_i = \mathbf{z}_i$.

The least-squares dummy variable (LSDV) estimator of the CEO effects is

$$\hat{\mathbf{z}}_i = \mathbf{P} \mathbf{y}_i - \hat{\alpha}_i = \mathbf{z}_i + \mathbf{P} \mathbf{e}_i. \quad (\text{A4})$$

The estimated CEO effects are the true effects plus the within-spell mean of shocks. (For our method, it does not matter how firm fixed effects $\hat{\alpha}_i$ are estimated, as we difference out firm means in all contrasts.)

6.2 Population Moments and Bias Terms

The relevant moments are

$$\mathbb{E}[\hat{\mathbf{z}}_i] = \mathbf{z}_i, \quad (\text{A5})$$

$$\mathbb{E}[\hat{\mathbf{z}}_i \hat{\mathbf{z}}_i'] = \mathbb{E}[(\mathbf{z}_i + \mathbf{P} \mathbf{e}_i)(\mathbf{z}_i + \mathbf{P} \mathbf{e}_i)'] = \Lambda_i + \sigma \mathbf{P} \Sigma \mathbf{P}, \quad (\text{A6})$$

$$\mathbb{E}[\hat{\mathbf{z}}_i \mathbf{e}_i'] = \mathbb{E}[(\mathbf{z}_i + \mathbf{P} \mathbf{e}_i) \mathbf{e}_i'] = \sigma \mathbf{P} \Sigma, \quad (\text{A7})$$

$$\mathbb{E}[\hat{\mathbf{z}}_i \mathbf{y}_i'] = \mathbb{E}[(\mathbf{z}_i + \mathbf{P} \mathbf{e}_i)(\mathbf{z}_i + \mathbf{e}_i)'] = \Lambda_i + \sigma \mathbf{P} \Sigma. \quad (\text{A8})$$

These show variance inflation by $\sigma \mathbf{P} \Sigma \mathbf{P}$ and contaminated covariances.

6.3 Bias in Regression Slopes

For any linear contrast with weights $\mathbf{w} \in \mathbb{R}^T$ and $\mathbf{w}' \mathbf{1} = 0$, define the estimated effect for the contrast by

$$\Delta \hat{\mathbf{z}}_i = \mathbf{w}' \hat{\mathbf{z}}_i = \mathbf{w}' \mathbf{P} \mathbf{y}_i = \Delta \mathbf{z}_i + \eta_i, \quad \eta_i := \mathbf{w}' \mathbf{P} \mathbf{e}_i,$$

and recall that $\Delta y_i = \Delta \mathbf{z}_i + \varepsilon_i$ where $\varepsilon_i = \mathbf{w}' \mathbf{e}_i$. For the adjacent-spell contrast used in the main text, η_i corresponds to $\Delta \bar{e}_{is}^k$.

The naive regression slope is

$$\begin{aligned} \hat{\beta} &= \frac{\text{Cov}(\Delta \hat{\mathbf{z}}_i, \Delta y_i)}{\text{Var}(\Delta \hat{\mathbf{z}}_i)} \\ &= \frac{\mathbf{w}' \mathbb{E}[\hat{\mathbf{z}}_i \mathbf{y}_i'] \mathbf{w}}{\mathbf{w}' \mathbb{E}[\hat{\mathbf{z}}_i \hat{\mathbf{z}}_i'] \mathbf{w}} \\ &= \frac{\mathbf{w}' [\Lambda_i + \sigma \mathbf{P} \Sigma] \mathbf{w}}{\mathbf{w}' [\Lambda_i + \sigma \mathbf{P} \Sigma \mathbf{P}] \mathbf{w}} \\ &= \frac{\lambda_i + \sigma \mathbf{w}' \mathbf{P} \Sigma \mathbf{w}}{\lambda_i + \sigma \mathbf{w}' \mathbf{P} \Sigma \mathbf{P} \mathbf{w}}. \end{aligned} \quad (\text{A9})$$

Covariance bias term. $\text{Cov}(\varepsilon_i, \eta_i)$ inflates the numerator when shocks are persistent and spells short.

Variance bias term. $\text{Var}(\eta_i) > 0$ attenuates the denominator.

Both biases vanish as $T_1, T_2 \rightarrow \infty$.

6.4 Sample Moments Across Heterogeneous Groups

With G groups of sizes N_g , sample moments are

$$\widehat{\text{Cov}}(\Delta y_i, \Delta \hat{\mathbf{z}}_i) = \sum_{g=1}^G \frac{N_g}{N} \widehat{\text{Cov}}_g(\Delta y_i, \Delta \hat{\mathbf{z}}_i) \quad (\text{A10})$$

$$\widehat{\text{Var}}(\Delta \hat{\mathbf{z}}_i) = \sum_{g=1}^G \frac{N_g}{N} \widehat{\text{Var}}_g(\Delta \hat{\mathbf{z}}_i) \quad (\text{A11})$$

The expected values are

$$\begin{aligned}\mathbb{E}\widehat{\text{Cov}}(\Delta y_i, \Delta \hat{z}_i) &= \sum_{g=1}^G \frac{N_g}{N} \left[\bar{\lambda}_g + \sigma_g \mathbf{w}_g' \mathbf{P}_g \Sigma \mathbf{w}_g \right] \\ &= \bar{\lambda} + \sum_{g=1}^G \frac{N_g}{N} \sigma_g \mathbf{w}_g' \mathbf{P}_g \Sigma \mathbf{w}_g\end{aligned}\quad (\text{A12})$$

$$\begin{aligned}\mathbb{E}\widehat{\text{Var}}(\Delta \hat{z}_i) &= \sum_{g=1}^G \frac{N_g}{N} \left[\bar{\lambda}_g + \sigma_g \mathbf{w}_g' \mathbf{P}_g \Sigma \mathbf{P}_g \mathbf{w}_g \right] \\ &= \bar{\lambda} + \sum_{g=1}^G \frac{N_g}{N} \sigma_g \mathbf{w}_g' \mathbf{P}_g \Sigma \mathbf{P}_g \mathbf{w}_g\end{aligned}\quad (\text{A13})$$

Here $\bar{\lambda}$ is the average CEO effect variance. Bias terms depend on $\mathbf{P}_g$, $\mathbf{w}_g$, and $\sigma_g \Sigma$, making direct estimation impractical with complex Σ .

APPENDIX B: ADDITIONAL FIGURES AND TABLES

Table B1: Company-Year Characteristics

Year	Firms	CEO Switches	Employment
1995	76866	3921	13
2000	133686	10043	10
2005	145956	8772	9
2010	138815	8684	9
2015	158242	10733	9
2020	174683	10714	9
Total	4107611	262973	10

Notes: the table presents the number of firms, number of CEO replacements, and average employment of firms in the data in various years.

Table B2: Industrial Distribution of Firms

Industry	N	%
Agriculture	141867	3.5
Industry	578437	14.1
Construction	447245	10.9
Trade	1122181	27.3
Business Services	662333	16.1
Other Services	1155548	28.1

Notes: Distribution of firms by sector.

Table B3: CEO Tenure

Statistic	P25	P50	P75
1st CEO tenure	2	4	9
2nd CEO tenure	2	4	7
3rd CEO tenure	2	3	6
4th or higher CEO tenure	1	3	5

Notes: distribution of the length of tenure of CEOs. Tenure measured in years.